

# When Self-Modification Becomes Memory

## An Audited Comparison of Three Retention-Interface Configurations Under a Shared Reflection Layer

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### Abstract

Language-model agents can edit prompts, memories, tools, and code, but self-improvement bundles five evidentiary propositions: retention, expression, behavioral mediation, task improvement, and recursive improvement. This study measures retention, expression, and task improvement systematically and audits behavioral mediation in three selected, trace-dependent cases. Recursive improvement requires descendant-productivity evidence and is not measured. We study one development lineage for each of three substrate-proposal-acceptance-interface configurations under a shared outer agent, 28-task curriculum, and 19-task cold-restart evaluation. The nominal state-hidden ablation was actor-visible-equivalent to cold, so the frozen primary comparison is one evolved lineage against one labeled draw from the no-context distribution, not a matched-lineage intervention. None of the three lineages met the written uplift rule, and the design cannot estimate a population null. Continuous-approximation planning differences were 30.7, 45.1, and 28.7 percentage points for SWE, Terminal, and ActiveGraph; effects smaller than these coarse values could not be reliably detected by this design. Six byte-identical initial-request no-context executions differed on 7 of 19 tasks. The sham-benefit conjunct is unevaluable because token load differed for every configuration and Hybrid also had a byte mismatch. The shared interface expressed retained structure mainly as text and exposed no retained invocation handle. Mechanism audits recorded policy-cap saturation, a common proposal-form bottleneck, and vocabulary-associated selection of failure-derived lessons. Two outcome-blind model-coding passes agreed that all 84 proposals shared one nonexecutable textual genre; one coder also generated the proposals, invariant-field kappa is undefined, and interpretive labels remain exploratory. We contribute an expression profile, an equivalent duplicates protocol, and a control ladder. The findings are descriptive and specific to this harness.

## 1 Introduction

Language-model agents now edit their own prompts, memories, tools, and source code. Prior systems study recursive improvers, runtime self-modification, coding-agent source changes, branching archives, and descendant-oriented search. These systems make self-modification

easy to observe. Self-improvement is a different empirical claim. A visible edit establishes that the system changed itself. It does not establish that the change was expressed at evaluation time, altered behavior, caused better externally graded outcomes, or improved the system’s capacity to produce useful future changes. Section 9 uses this literature only as conceptual context; comparative feature and performance claims require an independent primary-source audit before external circulation.

We separate the statement that an agent improved itself into five constructs: (1) **retention**, meaning persistent state changed; (2) **expression**, meaning retained state reached the evaluation-time policy; (3) **behavioral mediation**, meaning retained state causally changed the policy’s trajectory or resource use; (4) **task improvement**, meaning externally graded held-out performance rose beyond matched controls and stochastic spread; and (5) **recursive improvement**. These are distinct evidentiary propositions, not ordinal performance levels. Retention and actor-visible expression can be established descriptively; behavioral mediation, task improvement, and recursive improvement generally require counterfactual or lineage contrasts. This study systematically measures retention, expression, and task improvement. It records descriptive behavioral differences across all nine configuration-family cells and audits three selected complete-chain traces; those observations do not establish a general causal mediation effect. Descendant productivity is the required operationalization of recursive improvement and is not measured here.

We study those links in a heavily instrumented comparison of three persistent self-modification substrates chosen for the breadth of their native update surfaces: an arbitrary rewritable workspace, a success-gated procedure store, and a governed, hash-pinned executable ActiveGraph Pack. Each traversed the same ordered 28-task development curriculum and then faced 19 disjoint held-out tasks across three families after a cold restart. The design used external graders, four arms, immutable receipts, hash-bound adjudication, and a frozen success criterion. It also had one development lineage per configuration, one model, and small task families. The orchestration and local ActiveGraph family were self-built; SWE and Terminal used external official graders, and every empirical finding is scoped to the recorded harness. The frozen result is null: none of the three recorded lineages met the criterion. This paper explains why that null is informative

and why several favorable and unfavorable results inside our data dissolve under audit.

The audit measures mechanisms that recur across agent systems. First, budgeted context is a common route from memory to later action. Under that route, our three substrate-interface configurations delivered between 12% and 100% of their declared retained items per task, exposed no retained invocation handle in the actor-visible action schema, and expressed native structural change to the acting policy almost entirely as text. These percentages are within-substrate coverage measures rather than comparable quantities of useful information. Second, single-run task estimates can conceal substantial outcome variability. Our six no-context labels issued byte-identical first requests per task, seed included, and still differed on 7 of 19 tasks. Their family-level means spanned 2/10 tasks (20.0 points) on SWE and 1/6 task (16.7 points) on Terminal; the continuous-score ActiveGraph span was 21.3 points. Third, placebo context can change the computation it is meant to control. On SWE, the Workspace evolved arm solved 8/10 tasks and the sham solved 6/10, creating a 20-point evolved-minus-sham contrast; the six equivalent no-context labels on the same ten tasks already ranged from 6/10 to 8/10. Fourth, lexical retrieval and append-only retention created observable selection and capacity effects. A top-three retriever over-selected broad-vocabulary failure lessons, and the Pack eventually blocked three consecutive updates at its policy limit.

The shared reflection layer created a proposal-form bottleneck before configuration-specific acceptance and native mutation. Two separately executed outcome-blind model-coding passes using the recorded model identifiers `gpt-5.6-sol` and `gpt-5.6-terra` over all 84 update proposals assigned every proposal to the same representational genre: natural-language, append-only, nonexecutable, operational guidance with no specified activation. Interpretive fields were less stable across coders and remain exploratory pending human coding. One coder, Sol, was also the model that generated the proposals; Terra supplied the second pass. The compact package records distinct model identifiers and does not establish provider or pretraining independence between them. The cross-model consistency finding concerns the proposal layer. Native Workspace edits and Pack source changes remained structurally real and heterogeneous.

The receipts cut in both directions. An apparent transfer success involved an evolved SymPy pass matched by a no-context run that produced essentially the same fix, while the retained SymPy lesson concerned a different mechanism. An apparent retrieval failure scored exactly at the minimum of six equivalent no-context runs. The one evolved result outside its task’s exact equivalent range was a regression in which retained context accompanied a larger invalid intervention and the ablation produced the smaller correct fix. A tie-aware conditional exchangeability calculation expected 12/7 (1.71) outside-range results and observed one, so the case is a mechanistic trace rather than a population or chance detection. A case audit that

can show when favorable and unfavorable causal stories are unsupported is a minimum requirement for interpreting individual trajectories.

Prior systems motivate several distinct optimization and memory questions. This review round did not independently audit their feature coverage or reported performance, so the manuscript does not use a comparative matrix or quantitative prior-system claims as novelty evidence. Applying the present instruments in independently audited systems is a concrete next test.

This paper contributes six instruments and observations specified for reuse:

1. a five-construct decomposition separating retention, expression, behavioral mediation, task improvement, and recursive improvement;
2. an **expression profile** measuring storage reach, eligible-payload reach, semantic-unit reach, selection adaptivity, and execution reach;
3. an **equivalent duplicates** protocol that creates task-level empirical dispersion references from executions with byte-identical initial requests;
4. a **control ladder** organizing eight control designs by the nuisance they address and the claims they license;
5. mechanism checks for **vocabulary-mediated retrieval concentration**, source-policy saturation, and common-template reflection-proposal homogenization; and
6. a receipt-bound compact release with graders, requests, artifacts, and an adjudication ledger; complete trajectory traces remain in a separately gated archive pending disclosure review and deposit.

The scope is a within-study diagnostic demonstration. It does not supply a population estimate, a universal architecture ranking, or a universal auditability-expressivity law. The expression bottleneck was created by the implemented common interface, and that fact is central to the result. The reflection template determined what could be proposed, the acceptance gate determined what survived, the retriever and adapter determined what was expressed, and the execution stack determined what a score could mean. The interface and the acceptance mechanism are part of the self-improving system.

The contribution status is explicit:

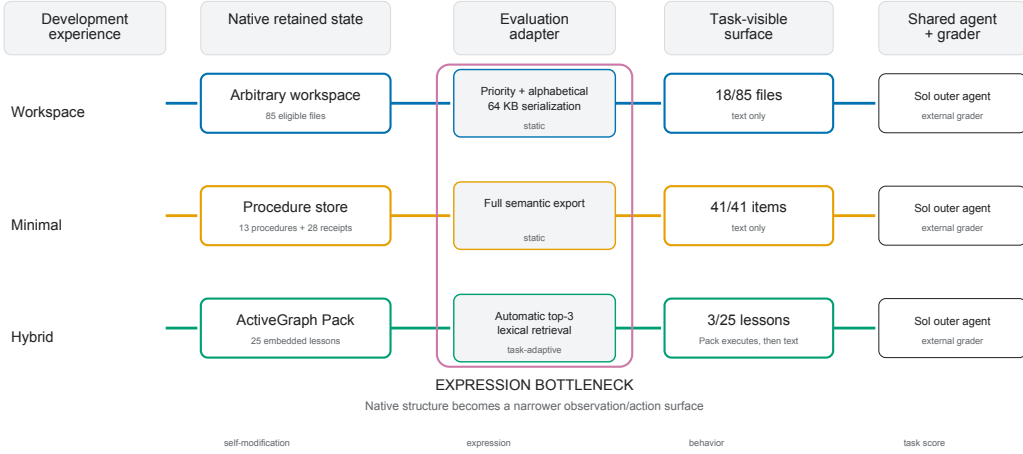
## 2 Measurement framework

Let  $R_i^t$  denote retained state for configuration  $i$  after development generation  $t$ . Let  $A_i(R_i^t, x)$  be the evaluation adapter for task  $x$ ,  $O_i^t(x)$  the observation and action surface emitted by the adapter,  $\pi(O_i^t(x), x)$  the induced acting policy,  $Y_i^t(x)$  the external task score, and  $M_i^t$  the productivity of the process that creates useful descendants. The five constructs correspond to different comparisons:

1. Retention:  $R_i^{t+1} \neq R_i^t$ .
2. Expression:  $O_i^{t+1}(x) \neq O_i^t(x)$ .
3. Behavioral mediation: retained state causally changes the induced policy trajectory or resource use.

Figure 1. The expression bottleneck

Persistent change is filtered before it can affect action; each arrow is a distinct empirical claim.



**Figure 1: The expression bottleneck.** The study separates retention, expression, behavioral mediation, task improvement, and recursive improvement. Retention is necessary but does not identify later score improvement or recursive self-improvement. The bold bottleneck is the implemented adapter boundary. It does not imply that all auditable interfaces reduce expressivity.

**Contribution status.** Systematic measurement is separated from selected trace audits and proposed endpoints.

| Proposition           | Status                                                 | Recorded evidence                                               | Boundary                                                  |
|-----------------------|--------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------|
| Retention             | Systematically exercised                               | Immutable state generations and receipts for all three lineages | Persistent change only                                    |
| Expression            | Systematically measured                                | Frozen adapter emissions and held-out semantic-unit coverage    | Actor-visible availability only                           |
| Behavioral mediation  | Descriptive summaries plus three selected trace audits | Model/tool/token summaries and complete-chain cases             | No general causal mediation effect; complete traces gated |
| Task improvement      | Systematically measured                                | 57 configuration-task outcomes and nine family-level contrasts  | One lineage per configuration; descriptive intervals      |
| Recursive improvement | Proposed external endpoint                             | Descendant productivity specified as the operationalization     | Not measured                                              |

4. Task improvement: expected external score increases beyond matched controls and relevant outcome spread.
5. Recursive improvement: the productivity  $M_i^t$  of generating useful descendants increases.

This decomposition prevents evidence from moving silently between levels. The five propositions have different logical status and need not move together. A new file establishes retention. A serialized lesson establishes possible expression. A changed trace establishes behavioral mediation. An external score contrast supports task improvement. Recursive improvement is the construct; descendant productivity is its operationalization. It requires evidence that earlier changes improved later update production.

The title’s threshold is operational rather than metaphysical: a self-modification becomes memory in this study when it survives restart and is eligible to reach a later policy through the recorded adapter. That threshold combines retention with actor-visible expression. It does not establish behavioral mediation, task improvement, or recursive improvement.

## 2.1 Expression profile

A single byte ratio cannot compare heterogeneous stores. Minimal’s complete SQLite state is mostly provenance, while its semantic export is small. Workspace bytes mix source, tests, instructions, and records. Hybrid executes a Pack query before the actor, so direct byte delivery omits runtime activation. We therefore report five structured axes:

1. **storage reach:** unique bytes copied into the initial actor-visible context divided by bytes in a declared canonical serialization of the complete frozen retained-state snapshot;
2. **eligible-payload reach:** those copied bytes divided by the canonical retained byte spans admitted by the adapter’s frozen eligibility rule;
3. **semantic-unit reach:** declared substrate-native retained items surfaced per task, plus their union across the probe, divided by the frozen item inventory;
4. **selection adaptivity:** whether selection is explicitly conditioned on task input, accompanied by the per-task-to-probe-union coverage expansion; and
5. **execution reach:** whether retained code runs automatically and how many retained invocation handles appear in the actor-visible action schema.

Semantic units are files for Workspace, typed procedures, capabilities, or receipts for Minimal, and lessons for Hybrid. The unitization rule and item types are frozen before evaluation. A Workspace file counts once when any bounded prefix of that file reaches the actor; completeness is reported separately. These ratios measure coverage within one substrate and must not be used to compare quantities of useful information across substrates. The coverage-expansion gap measures selection turnover; it does not by itself show that selection is relevant or beneficial. Interfaces without byte-copy semantics should report the byte axes as not applicable and declare a proxy rather than force a ratio.

The expression profile is diagnostic rather than ordinal. Greater reach can increase useful transfer, interference, cost, or all three.

## 2.2 Control ladder

The control ladder organizes controls by the nuisance they address and the inference they license. Rungs 3 through 6 are alternative payload controls, not a monotonic sequence in which every higher number dominates every lower number.

The present study separably implements rungs 1, 3, and 7. It records the lineage state hash required by rung 2, but the cold-ablation adapters validate and then suppress that state without exposing a lineage-derived channel to the actor. Cold and cold ablation therefore collapse at the actor-visible intervention boundary. Rung 2 is not separably implemented here. The other rows are operational specifications for follow-up designs rather than empirical controls claimed here.

## 3 Study design

All three configurations used the same `gpt-5.6-sol` outer coding agent, brokered tool protocol, task ceilings, grader boundary, and common reflection step. Accordingly, the experiment compares three proposal, acceptance, retention, and interface configurations downstream of one shared textual proposal generator. It does not compare unconstrained native self-modification processes. The configurations differed in retained state:

- **Workspace v1.2** could rewrite and retain an arbitrary workspace tree.
- **Minimal v2** stored every scalar receipt and promoted only successful experiences as procedures or deterministic capabilities. This recorded lineage produced 13 procedures and zero deterministic capabilities, so the capability-execution path was never exercised.
- **Hybrid Packs** used a model-authored, tested, hash-pinned ActiveGraph Pack to store and retrieve lessons.

The native mutation units differ in breadth. The score comparison concerns their implemented common interface and does not order their native ceilings.

Each configuration traversed the same 28-task devel-

opment curriculum: 20 SWE-bench Verified issues, six Terminal-Bench 2 tasks, and two local Ouro-ActiveGraph-50 systems. Evaluation used 19 disjoint held-out tasks: ten SWE-bench Verified issues, six Terminal-Bench 2 tasks, and three Ouro-ActiveGraph-50 systems. SWE and Terminal used official binary verifiers. The ActiveGraph family used normalized scores from 50 sealed checks. The local benchmark and the Hybrid Pack both use ActiveGraph technology, creating a structural author-built overlap. The sealed checks and the unfavorable Hybrid ActiveGraph result, -10.0 points versus its labeled no-context draw, mitigate simple directional bias but do not remove that conflict.

For each configuration and held-out task, the study ran four arms:

- **evolved**, with retained development state enabled;
- **cold ablation**, with the same lineage hash validated and learned state hidden;
- **sham**, with a seeded opaque artifact intended to approximate the evolved context size; and
- **cold**, with no development state.

Source inspection after Round 1 confirmed that cold and cold ablation have the same task, model, seed field, tools, workspace, system instructions, empty retained-context field, and later actor-visible tool protocol. The hidden lineage hash is validated for cold ablation but never reaches the actor. The frozen primary estimand, paired **evolved - cold ablation**, is therefore an evolved contrast against one labeled draw from the six actor-visible-equivalent no-context arms rather than a separable matched-lineage intervention. We keep that contrast as the frozen decision rule and report pooled-six and leave-the-matched-ablation-out sensitivities in Section 4.

The written frozen rule required benefit over the labeled cold-ablation draw and sham, a nominal 95% task-resampling interval above zero in at least two families including SWE or Terminal, valid cold reload, and no material development regression. This was an operational decision rule for the recorded probe rather than a claim of population-level interval coverage. The later source audit makes the labeled ablation a no-context draw, and the observed sham token and byte mismatches make the sham-benefit conjunct unevaluable. These limitations cannot convert a failed written rule into evidence of uplift.

The information boundary excluded hidden tests, expected outputs, private cases, grader logs, and oracle artifacts from lineage state. Each accepted generation produced an immutable directory whose manifest bound architecture, parent run, development experience, artifact size, and hash. Evaluation occurred after cold reload. The study specification is `../research/local_study.json`, SHA-256 `8313d21971452704893200d2022718feca985142ae32fe7d16932d34a0e0d581`, committed with frozen status at study commit `559843894578141dfbd39b7bae2246db76e57bde` on 2026-07-17T18:30:38-07:00. The compact release does not preserve an independently witnessed timestamp for the first development call, so we

**Control ladder.** Payload controls 3–6 are alternatives rather than a monotonic sequence.

| Control                       | Intervention and held-fixed variables                                                                               | Licensed inference                                                            |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 1. Cold floor                 | Same model, task, harness, and budget; no lineage                                                                   | Model-and-harness floor                                                       |
| 2. State-hidden ablation      | Same developed lineage and evaluation stack; learned state hidden; a distinct actor-visible channel is required     | Effect of enabling retained state only when that distinct channel is verified |
| 3. Opaque byte-size sham      | Seeded opaque text; report byte ratio and token load                                                                | Sensitivity to opaque payload of recorded size                                |
| 4. Token-matched neutral sham | Match tokenizer, position, and count with a preregistered corpus and blinded relevance screen                       | Payload effect after equalizing first-call token load                         |
| 5. Semantic sham              | Plausible units from disjoint task families under blinded relevance screening                                       | Sensitivity to screened-unrelated guidance                                    |
| 6. Structure-preserving sham  | Preserve count, schema, and token budget while permuting content or metadata                                        | Sensitivity to retained structure apart from its original mapping             |
| 7. Equivalent duplicates      | Byte-identical initial request, model, seed field, task image, harness, and budgets; later trajectories may diverge | Observed within-task execution dispersion                                     |
| 8. Native-interface ablation  | Hold state, task, model, and budgets fixed while toggling native execution                                          | Effect of the native execution path with recorded downstream changes          |

call this a frozen study specification and decision rule rather than an externally registered preregistration. The manuscript package reviewed in Round 1 was commit 8bdf502cc08be968b21bbe5ebf177d49879a0fb.

The study completed 84 development attempts and 228 held-out attempts. Failed attempts remained in the denominator. Six SWE runs reached the frozen 3,600-second task limit. Six Terminal agents reached the 1,800-second limit, and completed verifiers returned zero. One ActiveGraph grader contradicted a receipt-bound parsable submission; one sealed grader-only retry of the hash-identical artifact recovered 24/50. No model attempt was rerun and no raw row was rewritten. The adjudication index binds the raw report, datasets, changed analytical row, and evidence by SHA-256.

Scores remain separate by family. Nominal 95% descriptive task-resampling intervals use 10,000 deterministic percentile-bootstrap draws. Each draw resamples the paired per-task arm differences jointly with replacement. With one lineage and ten, six, and three held-out tasks, no frequentist coverage is asserted. The three-task ActiveGraph interval is especially discrete and unstable and is not displayed in the headline outcome table.

The compact package records the outer model identifier and the byte-identical initial requests. It does not record a provider guarantee of deterministic sampling, endpoint immutability, temperature, top-p, or context-window settings. The seed field is therefore a best-effort request parameter rather than a determinism guarantee.

## 4 Frozen outcome rule and pooled-control sensitivity

None of the three recorded lineages met the frozen uplift criterion. This is a descriptive result from one development lineage per configuration. The labeled comparator is one draw from the actor-visible-equivalent no-context distribution, not a separable matched-lineage intervention, and the study cannot estimate a population null. Table 1 reports the binary families as fractions and percentages. For context, the six equivalent no-context labels ranged from 6/10 to 8/10 on SWE and from 4/6 to 5/6 on Termini-

nal despite byte-identical initial requests within each task. The continuous-approximation planning differences were 30.7 percentage points for SWE, 45.1 for Terminal, and 28.7 for ActiveGraph. Effects smaller than these coarse, descriptive values could not be reliably detected by this design; the values are not decision thresholds.

Intervals equal to  $[0, 0]$  are degenerate because every paired task difference was zero; they do not imply population-level certainty. The labeled no-context column preserves the frozen comparison even though source inspection later showed that it was one draw rather than a separable lineage intervention. The sham columns are descriptive only: sham first-call token load was 1.88 to 2.43 times the incremental evolved token load per byte, and Hybrid’s sham was only 56% to 58% of mean evolved context bytes. The sham conjunct of the written rule is therefore formally unevaluable and supplies no benefit-control decision in this probe.

The pooled-six sensitivity recomputes the descriptive mean contrast against all actor-visible-equivalent no-context labels. In Workspace, Minimal, and Hybrid order, the evolved-minus-pooled deltas are +6.7, -3.3, and +6.7 points for SWE; +11.1, -22.2, and -5.6 points for Terminal; and -9.9, +4.1, and -9.9 points for ActiveGraph. Leaving each configuration’s labeled draw out of its reference gives +8.0, -6.0, and +6.0; +10.0, -20.0, and -6.7; and -8.4, +3.2, and -9.9 points, respectively.

The corresponding nominal 95% task-resampling intervals, again in Workspace, Minimal, and Hybrid order, are as follows. For SWE, pooled-six intervals are [-8.3, +26.7], [-30.0, +18.3], and [-3.3, +21.7] points; leave-one-out intervals are [-10.0, +32.0], [-32.0, +16.0], and [-4.0, +20.0]. For Terminal, pooled-six intervals are [0.0, +27.8], [-50.0, 0.0], and [-25.0, +8.3]; leave-one-out intervals are [0.0, +30.0], [-46.7, 0.0], and [-30.0, +10.0]. For ActiveGraph, pooled-six intervals are [-33.3, +3.7], [-6.3, +18.7], and [-33.3, +3.7]; leave-one-out intervals are [-29.6, +4.4], [-5.6, +15.2], and [-32.0, +2.4]. No interval has a lower bound above zero. The sensitivities therefore do not produce the required positive pattern in two families, and the no-uplift decision is unchanged. The exact nine-cell calculation is `data/generated/equivalent-control-sensitivity.csv`.

**Table 1:** Frozen held-out outcomes. Binary-family intervals are nominal descriptive task-resampling intervals; ActiveGraph has only three tasks, so its intervals are not displayed. The sham columns are descriptive because token load was unmatched and Hybrid also had a byte mismatch.

| Family          | Configuration | Evolved | Labeled no-context draw | Evolved-draw [interval]   | Sham   | Evolved-sham |
|-----------------|---------------|---------|-------------------------|---------------------------|--------|--------------|
| SWE (10)        | Workspace     | 8/10    | 8/10                    | 0/10 [0, 0] pp            | 6/10   | +2/10        |
|                 | Minimal       | 7/10    | 6/10                    | +1/10 [-20, +40] pp       | 7/10   | 0/10         |
|                 | Hybrid        | 8/10    | 7/10                    | +1/10 [0, +30] pp         | 8/10   | 0/10         |
| Terminal (6)    | Workspace     | 5/6     | 4/6                     | +1/6 [0, +50] pp          | 3/6    | +2/6         |
|                 | Minimal       | 3/6     | 5/6                     | -2/6 [-66.7, 0] pp        | 4/6    | -1/6         |
|                 | Hybrid        | 4/6     | 4/6                     | 0/6 [0, 0] pp             | 4/6    | 0/6          |
| ActiveGraph (3) | Workspace     | 82.7%   | 100.0%                  | -17.3; interval not shown | 96.0%  | -13.3 pp     |
|                 | Minimal       | 96.7%   | 88.0%                   | +8.7; interval not shown  | 100.0% | -3.3 pp      |
|                 | Hybrid        | 82.7%   | 92.7%                   | -10.0; interval not shown | 82.7%  | 0.0 pp       |

**Table 2:** Retained state after 28 development tasks.

| Substrate | Accepted after |             | Final retained product                     |
|-----------|----------------|-------------|--------------------------------------------|
|           | Passed task    | Failed task |                                            |
| Workspace | 13             | 12          | 429,731-byte workspace                     |
| Minimal   | 13             | 0           | 13 procedures, 0 capabilities, 28 receipts |
| Hybrid    | 15             | 10          | 25 lessons; 62,506-byte Pack               |

The table and sensitivity establish persistent evaluation after development and the absence of frozen uplift. They do not explain whether retained state was available, whether it changed behavior, or whether visible contrasts exceeded outcome variability. Sections 5 through 7 audit those links.

## 5 Expression bottleneck

All three substrates created cold-loadable state:

These receipts establish retention. The expression profile shows what could reach later behavior:

Minimal supplies an observed counterexample, in this implementation, to storage reach as a sufficient metric. Its database and provenance artifacts were large, giving 1.35% storage reach, while the adapter exported all declared semantic units. Those 41 units comprised 13 procedures and 28 receipts. None was a deterministic capability. The byte numerators and denominators are Workspace 64,000/429,731 storage and 64,000/247,411 eligible payload; Minimal 24,767/1,839,823 and 24,767/24,865; and Hybrid mean 10,677/266,265 and 10,677/62,319. For Minimal, the 98-byte difference is between the complete export file and the canonical serialization of its three admitted arrays; all 41 semantic units were present. For Hybrid, the 62,319-byte eligible denominator sums the 25 lesson payloads and excludes the Pack source wrapper, explaining its difference from the 62,506-byte final source size.

Workspace exposed 17 complete files and a bounded prefix of an eighteenth. Under the frozen semantic-unit rule, a file counts once when any bounded prefix reaches the actor, yielding 18/85; completeness is reported separately. All post-priority files came from the alphabetically early `_development/` directory, and the actor could not invoke retained tools. Minimal delivered the same complete export to every task. Hybrid alone selected different

learned units by task: three of 25 per task and 17 of 25 across the probe. Its Pack executed a retrieval before each task, while the actor received only the returned text and could not call the Pack interactively.

The common adapter therefore narrowed each native substrate in a different way. Workspace was truncated and static. Minimal was semantically complete and static. Hybrid used task-conditioned selection with 56 points of probe-union coverage expansion. No actor-visible action schema contained a retained invocation handle. Minimal had no retained executable unit, while Workspace and Hybrid retained executable material that the actor could not invoke on demand. This supports an expression bottleneck in the implemented interface. A general auditability-expressivity tradeoff remains a hypothesis for a future interface intervention.

## 6 Equivalent duplicates and control interference

The six no-context labels were Workspace cold, Workspace ablation, Minimal cold, Minimal ablation, Hybrid cold, and Hybrid ablation. For each task, their first model requests were byte-identical, including seed. Their outcomes varied on 7 of 19 tasks. The remaining 12 tasks were stable: ten fixed passes and two fixed failures. Later model sampling, tool outputs, timing, and trajectories were uncontrolled execution variation, but actor-visible task, model, seed field, tools, workspace, system instructions, retained-context field, and tool protocol were the same. Three of these six labels are also the labeled ablation comparators in Table 1, so the dispersion reference and frozen comparator are overlapping rather than independent samples.

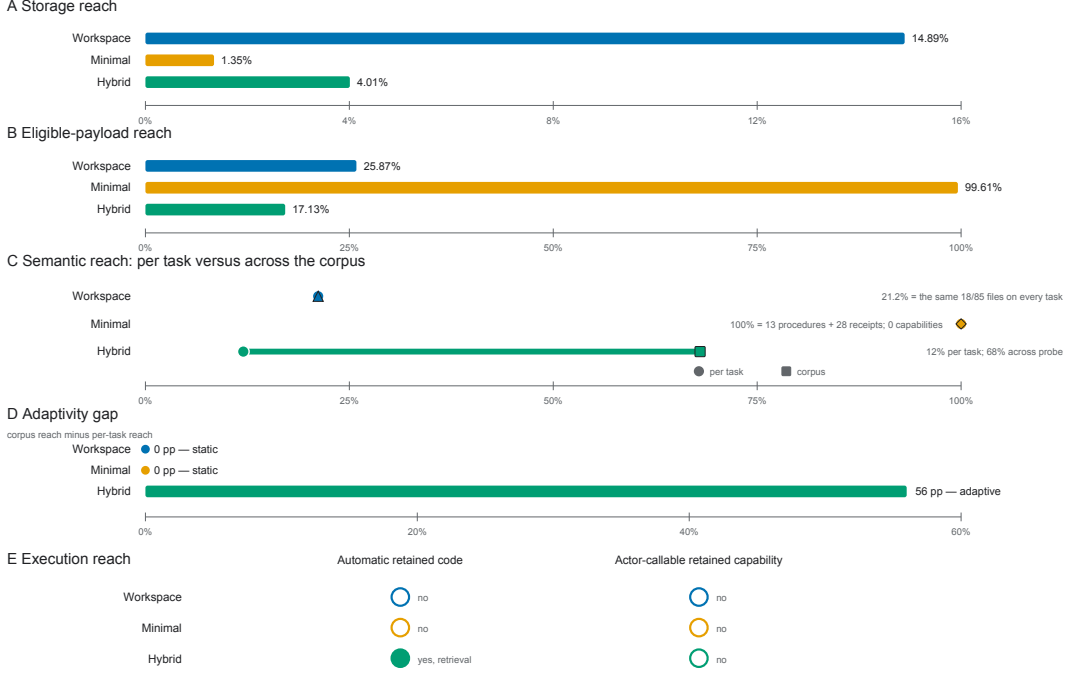
The one-way random-effects ICC treats tasks as targets and the six named no-context labels as exchangeable single replicate executions on the observed score scale. It is a descriptive repeatability coefficient, and the three-target ActiveGraph value is especially unstable. The design-size diagnostic is  $(1.96 + 0.842) \times \widehat{SE}$ , using a two-sided nominal  $\alpha = 0.05$ , 80% power, and the within-task variance across the six equivalent labels. It is a continuous normal approximation for a paired mean contrast. On the binary families it does not represent an attainable observed score increment. The three-task ActiveGraph value is illustrative only and must not be used to inter-

**Table 3:** Expression profile. Coverage ratios are within-substrate and do not compare useful information. Workspace counts a bounded partial-file prefix once and reports completeness separately.

| Substrate | Storage   | Eligible   | Semantic: task; union   | Selection; expansion             | Execution: automatic; handles   |
|-----------|-----------|------------|-------------------------|----------------------------------|---------------------------------|
| Workspace | 14.9%     | 25.9%      | 18/85 (21.2%); 18/85    | static; 0 pp                     | none; 0, retained tools hidden  |
| Minimal   | 1.35%     | 99.6%      | 41/41 (100%); 41/41     | static; 0 pp                     | none; N/A, 0 executable units   |
| Hybrid    | 4.0% mean | 17.1% mean | 3/25 (12%); 17/25 (68%) | task-conditioned; 56 pp turnover | automatic Pack query; 0 handles |

**Figure 2.** Expression profiles are multidimensional

Minimal’s 1.35% storage reach coexists with 100% semantic reach; no single byte ratio is sufficient.



**Figure 2: Expression is a profile.** Five axes are reported: storage reach, eligible-payload reach, semantic coverage with per-task and probe-union submeasures, selection adaptivity, and execution reach with automatic and actor-visible submeasures. Percentages are within-substrate coverage measures and do not compare quantities of useful information. Minimal’s 41 declared units are 13 procedures and 28 receipts; none is executable. Hybrid conditions top-three selection on task text, exposing three of 25 lessons per task and 17 across the probe. Its Pack runs automatically in the adapter and supplies no on-demand actor-visible invocation handle.

pret the continuous-score spread as a stable power estimate. Neither planning quantity is a decision threshold.

All discriminative variation in the no-context probe was concentrated in seven tasks that changed outcome under equivalent requests. Stable tasks still provide one-sided opportunities: fixed passes can expose regressions and fixed failures can expose gains. Minimal’s evolved failure on `scikit-learn__scikit-learn-13124`, against six equivalent passes, was the only one of 57 evolved configuration-task results outside an exact task-specific equivalent range. Under a tie-aware conditional exchangeability calculation, the expected count is 12/7 (1.71): 1/7 among 36 cases on stable control tasks and 11/7 among 21 cases on variable control tasks. The observed count was one and therefore supplies no positive chance detection. Method and family breakdown are recorded in `data/generated/outside-range-exchangeability.json`.

The maximum of the 15 absolute pairwise label-mean

gaps is reported for each family. The 15 values are dependent functions of six labels, and the labels are not randomized evolution replications. The maximum is a descriptive dispersion reference rather than a significance threshold.

The sham comparison exposes a second control problem. Workspace SWE evolved minus sham was +2/10 tasks (+20.0 points) because the evolved and ablation arms each solved 8/10 tasks while the opaque sham solved 6/10; equivalent no-context labels on the same tasks also ranged from 6/10 to 8/10. On Terminal, Workspace evolved minus sham was +2/6 tasks (+33.3 points) because evolved solved 5/6, ablation solved 4/6, and sham solved 3/6; equivalent labels on those tasks ranged from 4/6 to 5/6. The SWE sham family score fell within the equivalent-label range, at 6/10 against 6/10 to 8/10. The Terminal sham fell below all six equivalent family means, at 3/6 against 4/6 to 5/6. With six dependent draws, a

**Table 4:** Equivalent duplicate-execution dispersion and planning diagnostics. Neither planning quantity is a decision threshold.

| Family      | Tasks | Tasks variable across six labels | ICC(1,1)       | Maximum pair gap | Continuous-approx. planning |
|-------------|-------|----------------------------------|----------------|------------------|-----------------------------|
| SWE         | 10    | 3                                | 0.72           | 2/10 (20.0 pp)   | 30.7 pp                     |
| Terminal    | 6     | 2                                | 0.66           | 1/6 (16.7 pp)    | 45.1 pp                     |
| ActiveGraph | 3     | 2                                | 0.31, unstable | 21.3 pp          | 28.7 pp                     |

seventh observation below the observed minimum is consistent with interference and does not establish it. The sham consumed approximately 1.88 to 2.43 times as many incremental first-call tokens per byte as evolved context, depending on configuration. Its context bytes exactly matched evolved context for Workspace and Minimal. For Hybrid, the 6,093-byte sham was 56% to 58% of the mean evolved context across families. Byte matching therefore controlled artifact size while leaving tokenization, semantic load, and interference uncontrolled for Workspace and Minimal; Hybrid also retained a substantial byte-size mismatch.

Across all nine configuration-family cells, the recorded evolved execution also differed from the pooled no-context reference on at least one mean resource measure among model calls, tool calls, input tokens, output tokens, and cost. The complete descriptive summary is `data/generated/behavior-mediation.csv`. These systematic resource differences establish observed trajectory or resource divergence. They do not identify retained state as the causal mediator because the comparison is one lineage against overlapping execution labels and the later trajectories are stochastic. Section 8 therefore limits mechanism-level interpretation to three selected trace audits.

## 7 Mechanism audits

### 7.1 Vocabulary-mediated retrieval concentration

Hybrid retained 25 lessons, including 15 from passing development attempts and ten from failed attempts. Across 19 held-out tasks, the top-three retriever filled 57 slots. Failure-derived lessons occupied 37/57 slots (64.9%) despite forming 10/25 of the corpus (40%). Two failed Pylint lessons occupied 22/57 slots (38.6%). Only 17/25 lessons were ever retrieved, and 5/57 slots shared a repository prefix with the held-out task.

The implemented ranker counted set overlap between task words and an index containing suite, task identifier, title, scope, and trigger terms. Failure-derived lessons had broader indexed vocabularies. The raw enrichment, 37/57 retrieval slots from 10/25 failure-derived lessons, is the total association. Indexed vocabulary correlated with retrieval count within passing lessons ( $\rho = 0.743, n = 15$ ) and failed lessons ( $\rho = 0.763, n = 10$ ). In the descriptive regression `retrievals ~ indexed vocabulary + failure flag`, vocabulary had a coefficient of 0.194 retrievals per additional term and the adjusted failure co-

efficient was -0.093 retrievals. A 50,000-draw descriptive within-status label-permutation diagnostic shuffled vocabulary values within pass/fail strata, refit the vocabulary coefficient, and counted absolute coefficients at least as large as observed with a plus-one Monte Carlo correction. It gave  $p_{MC} \approx 0.0045$ . This conditional reference requires exchangeability of vocabulary labels within status; the 25 lessons form one fixed corpus and compete for 57 retrieval slots, so the value is neither causal nor population inference.

Offline re-ranking the same prompts changed the selection pattern without new model calls. Failure slots fell from 37/57 under raw overlap to 33/57 under Jaccard, 27/57 under binary cosine, and 28/57 under BM25. BM25 surfaced 22/25 lessons and reduced the top-two share from 22/57 to 10/57. These diagnostics are consistent with vocabulary-mediated selection in the implemented overlap ranker. They do not identify whether failure status has additional effects outside that path and do not show that alternative retrieval improves downstream task scores.

### 7.2 Capacity saturation

Hybrid Pack source grew from 7,053 to 62,506 bytes. Generation 25 first exceeded the 64,000-byte policy and succeeded after one repair. Every candidate in generations 26 through 28 remained over the cap after four author or repair requests; the closest final candidate was 65,928 bytes. Mean authoring requests rose from 1.0 over the first five generations to 3.2 over the last five, and mean authoring cost rose from \$0.249 to \$2.114.

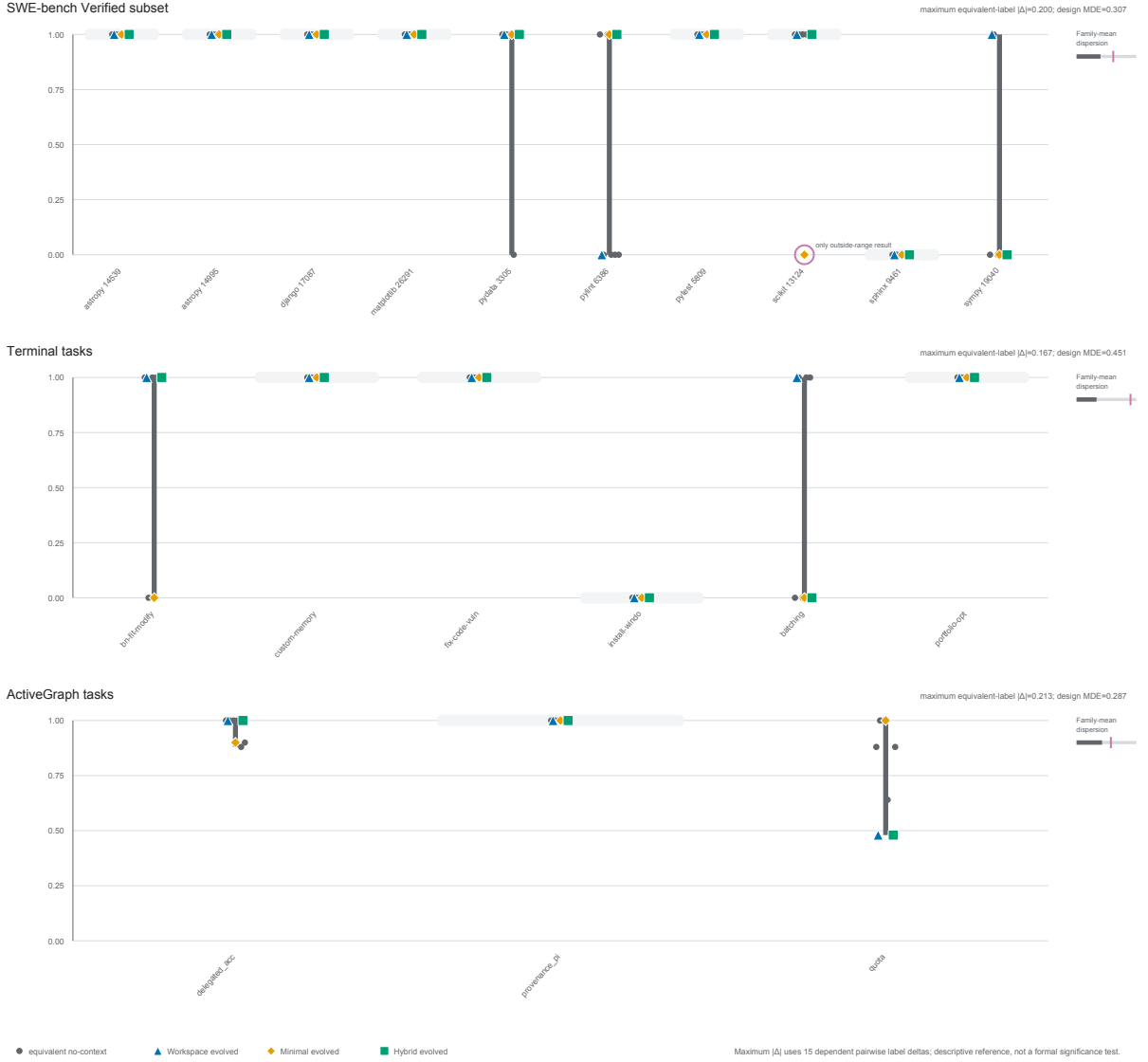
The policy rejection and retained-state plateau record source-policy saturation under the implemented 64,000-byte cap and append-oriented design. The cap was not varied, so this is not evidence of an inherent append-only limitation. Task identity and difficulty are confounded with generation order, so the cost trend is descriptive. The result does not establish a general limitation of ActiveGraph or executable memory.

### 7.3 Update-proposal taxonomy

A frozen codebook was applied to all 84 generation-level reflection proposals. Each coder saw opaque shuffled excerpts without architecture, task, sequence, score, acceptance, cost, or later retrieval metadata. `gpt-5.6-sol` and `gpt-5.6-terra` completed separately executed, outcome-blind coding passes. Sol also generated the proposals it later coded; Terra did not. The package records distinct identifiers and does not claim provider or pretraining independence. Original labels were hashed before comparison

Figure 3. Evolved outcomes versus task-specific duplicate-execution clouds

Six byte-identical no-context requests define each gray cloud; Minimal/scikit-learn is the only evolved result outside a cloud.



**Figure 3: Task-level initial-request duplicate spread.** For each of 19 tasks, gray points show six no-context outcomes generated from byte-identical initial model requests; colored marks show the three evolved outcomes. Later trajectory inputs may diverge. Seven tasks varied under the duplicate executions and 12 were stable. Minimal/scikit-learn is the only evolved configuration-task outcome outside the observed duplicate-execution range. The bar is the maximum of 15 dependent absolute pairwise label-mean gaps and is descriptive rather than a formal significance threshold. Neither the bar nor the planning diagnostic is a decision threshold.

and no disagreement was adjudicated.

Both coders labeled 84/84 proposals as natural-language, append-only, nonexecutable, operational guidance with no specified activation. They agreed on failure specificity for 83/84 proposals and validation strategy for 82/84. These replicated labels support a stable description of proposal form.

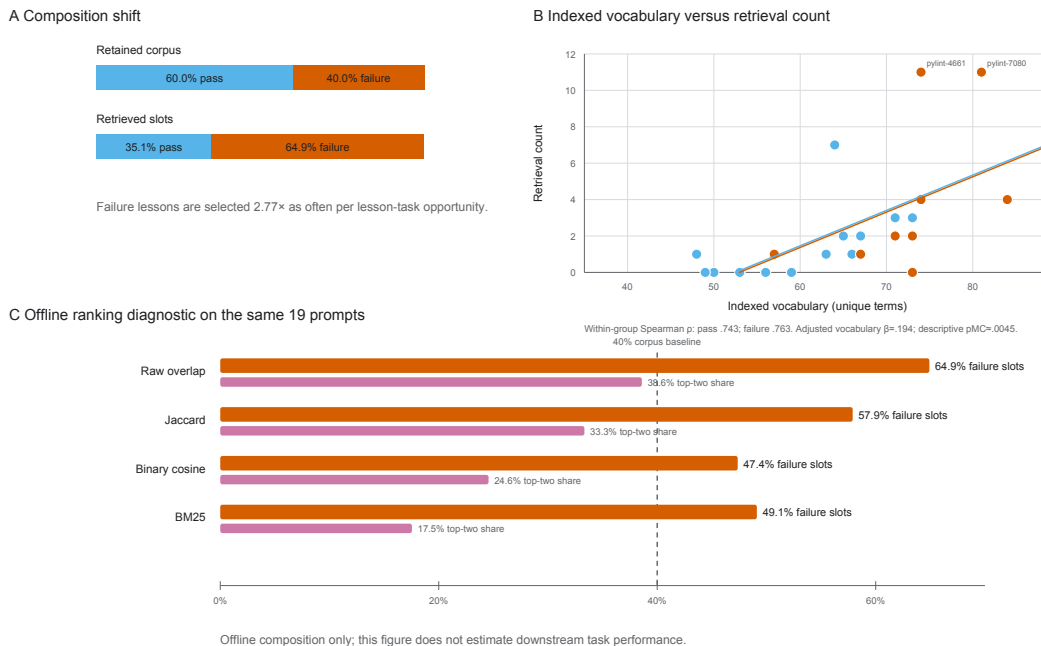
Interpretive fields were less stable and remain exploratory. Their complete agreement results and coder marginals appear in Appendix E.

The shared reflector therefore produced a homogeneous

representational genre. Configuration-specific gates selected different retained mixtures: Minimal accepted 13 success-derived proposals and no failure-derived proposals; Workspace accepted 25 proposals including 12 after failures; Hybrid accepted 25 including ten after failures. Human inter-rater reliability remains pending. The 84/84 form agreement is a cross-model consistency check, not independent human reliability. The analysis describes common update proposals rather than every native Workspace edit or Pack source change.

**Figure 4. Vocabulary-mediated retrieval concentration**

Failures are 40% of memory but 64.9% of retrievals; vocabulary predicts selection within both outcome groups.



**Figure 4: Vocabulary-mediated retrieval concentration.** Hybrid’s retained corpus was 40% failure-derived, while failures occupied 37/57 retrieval slots (64.9%). Retrieval count increased with indexed vocabulary within both success and failure groups. Because failure status plausibly acts through indexed vocabulary breadth, the adjusted regression is a mechanism diagnostic rather than an attribution of signal away from failure. It does not establish an effect on held-out score.

## 8 Audited cases

Case selection followed a frozen rule that required a complete chain:

retained update -> expression -> plausible mechanism -> later behavior

The rule also required comparison with the exact task-specific range among equivalent duplicates.

**Apparent success.** Workspace passed a held-out SymPy task. Its retained SymPy lesson concerned a different mechanism, and a no-context run generated essentially the same fix. Equivalent outcomes on the task ranged from failure to pass. The receipts establish a pass but do not support the learned-transfer story.

**Apparent failure.** Hybrid scored 0.48 on quota scheduler after retrieving mechanistically skewed lessons. Six equivalent no-context runs on the same task ranged from 0.48 to 1.00. The evolved score equaled the observed minimum. The receipts establish retrieval skew but do not identify that skew as the cause of the outcome.

**Outside-range regression.** Minimal failed a scikit-learn task that all six equivalent no-context runs passed. The retained-context trajectory changed an algorithm and rewrote an established expected-output test. The ablation made a smaller fix and passed the restored official test. This is consistent with harmful behavioral mediation in that trace. Across all 57 configuration-task cases,

however, one outside-range result was observed against a tie-aware conditional expectation of 1.71, so the case is not a population or chance detection. The trace does not identify one retained sentence as the cause.

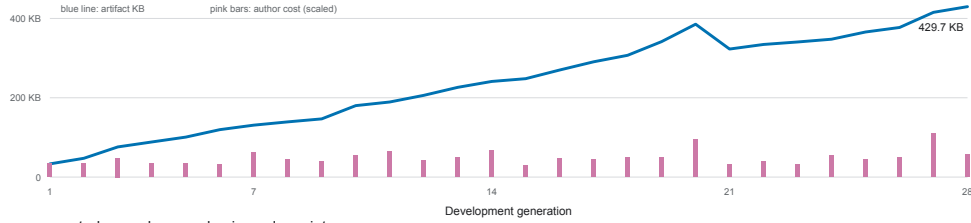
**Capacity ceiling.** Hybrid generation 25 succeeded after a repair near the policy cap. Generations 26 through 28 produced four over-cap candidates each. Under the frozen 64,000-byte source policy, the append-oriented source reached the cap and subsequent over-cap candidates were rejected. The case does not identify what would happen under another cap or a consolidation-enabled arm.

Together, the cases show why receipts must be permitted to invalidate stories in both directions. They also show why score, mechanism, and attribution should be reported separately.

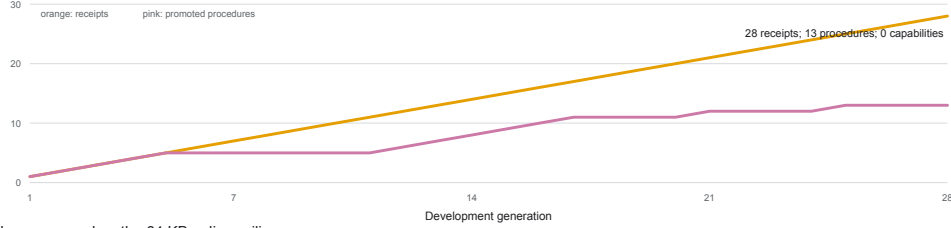
Figure 5. Accumulation, selective promotion, and saturation

All substrates retained state; only Hybrid encountered an explicit capacity ceiling.

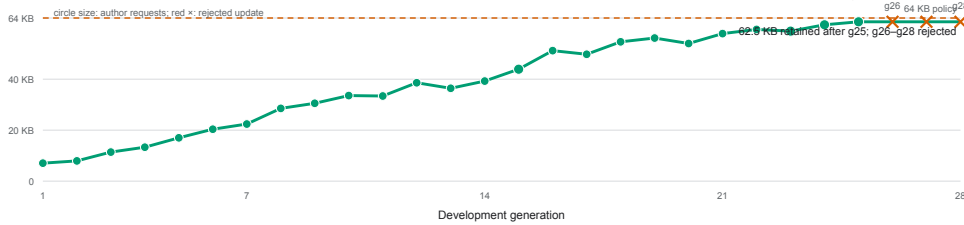
A Workspace: artifact growth and authoring cost



B Minimal: success-gated procedures and universal receipts



C Hybrid: Pack source reaches the 64 KB policy ceiling



**Figure 5: Accumulation, rising effort, and saturation.** Panel A shows Workspace artifact growth and authoring cost; panel B shows Minimal’s universal receipts and success-gated procedures; panel C shows Hybrid Pack growth and rejected updates at the 64,000-byte source policy. Hybrid generation 25 succeeded after repair and generations 26 through 28 produced no within-cap candidate. The cost trend is descriptive because generation order is confounded with task identity and difficulty.

## 9 Related work

Self-improving agent work spans recursive improvers, run-time self-modification, persistent coding-agent changes, branching search, and language-based memory. Examples include STOP [1], Gödel Agent [2], SICA [3], the Darwin Gödel Machine [4], the Huxley-Gödel Machine [5], and Reflexion [6]. Adjacent evaluation and memory work motivates attention to costs, holdouts, attribution, execution variability, acceptance noise, and consolidation.

The closest conceptual precedents come from three traditions. Memory evaluation distinguishes stored, retrieved, selected, and utilized material; causal mediation separates intervention, mediator, trajectory, and outcome; systems provenance binds events without by itself identifying causal effects. This paper combines those distinctions for persistent self-modification, profiles expression across heterogeneous retained substrates, keeps descendant productivity as a separate recursive endpoint, and binds within-study claims to receipts and actor-visible controls. It does not claim that the individual components are unprecedented.

This Round 3 revision removes the author-classified feature matrix and does not use quantitative prior-system performance claims. Those materials have not received

an independent primary-source audit. Before external circulation, a separate literature-verification gate must confirm the exact cited versions, the conceptual descriptions above, and any stronger comparison that might later be proposed. No such audit is part of the present study or revision.

## 10 Discussion

The frozen result does not establish that persistent self-modification cannot improve agents. It establishes that three retained configurations, under one common interface and one recorded lineage each, did not meet the specified uplift criterion when each was compared with one labeled draw from an actor-visible-equivalent no-context distribution. This is not a population null. The coarse planning differences of 30.7, 45.1, and 28.7 percentage points also show that smaller SWE, Terminal, and ActiveGraph effects could not be reliably detected by this design. The mechanism audit explains why a score-only summary would discard the most informative evidence.

First, native update breadth and evaluation-time expression are separate design dimensions. Workspace retained arbitrary files, Minimal retained procedures and receipts, and Hybrid retained executable Pack source. The actor received truncated static text, complete static text, or task-conditioned text. It received no callable retained capability. Evaluation therefore compared native substrates after a consequential common transformation.

Second, proposal, acceptance, retrieval, and expression form a pipeline. The common reflector produced one representational genre across all 84 proposals. Acceptance gates created different epistemic mixtures. Workspace and Hybrid retained failure-derived proposals that Minimal excluded. Hybrid’s lexical retriever then concentrated exposure on broad-vocabulary failure lessons. Each mechanism belongs inside the system boundary because each changes what a later actor can use.

Third, control labels can collapse or confound contrasts. Cold and the nominal state-hidden ablation preserved different provenance but delivered the same actor-visible intervention, so the frozen single-ablation comparison is one draw from the no-context distribution. Pooled-six and leave-one-ablation-out sensitivities retain intervals that include zero. The opaque sham matched evolved bytes for Workspace and Minimal but not Hybrid, and it differed in tokenization and semantic load. The frozen benefit-over-sham conjunct is therefore unevaluable. Equivalent duplicates varied despite identical first requests. A positive evolved-minus-sham contrast can combine useful expression, possible sham interference, and ordinary outcome spread. The control ladder makes these alternatives explicit and suggests the next diagnostic comparison.

Fourth, append-only retention carries a measurable maintenance burden. Hybrid reached a hard cap; Workspace and Minimal accumulated without consolidation. This observation motivates consolidation, pruning, and replacement as first-class update operations. It does not establish which operation would improve held-out performance.

The instruments are specified so they can be applied to persistent agents whose retained-state inventory, eligibility rule, adapter emissions, and actor-visible actions can be frozen and inspected. Systems without byte-copy semantics should declare a computable proxy or mark the

byte axes not applicable. Equivalent duplicates can be retained whenever multiple labels issue byte-identical initial requests under a common execution contract. The control ladder can be selected according to the intended causal claim. The update-proposal taxonomy can be applied to shared reflectors while native state changes are analyzed separately. Receipts can bind retained artifacts, exposures, trajectories, graders, and adjudication.

## 11 Limitations

The study completed one development lineage per configuration. The six equivalent labels expose outcome variability and do not replace independent evolution replications. All configurations shared one model and outer agent, which may dominate configuration differences. The held-out families contained ten, six, and three tasks. The three-task ActiveGraph family can demonstrate the instrument mechanics and report task-specific observations, but it cannot support a stable family-level effect estimate. Its uncertainty and planning diagnostics are especially unstable.

Cold and cold ablation were actor-visible-equivalent. Although cold ablation validated a lineage state hash, it provided no lineage-derived input, tool, workspace, instruction, or later protocol difference. The frozen primary contrast is therefore not a separable matched-lineage intervention. Pooled and leave-one-ablation-out sensitivities use the available equivalent controls but do not create independent evolution-lineage replication.

The common interface constrained native expression and exposed no retained interactive capability. This limits architecture ranking and creates the expression bottleneck examined in the paper. A general auditability-expressivity relationship requires an intervention that varies the interface while holding retained state fixed.

Minimal produced no deterministic capability in its recorded lineage. Its results therefore describe a procedure-and-receipt store with an unexercised capability path, rather than the ceiling of the broader design.

The sham targeted byte matching rather than token or semantic matching, and Hybrid retained a substantial byte-size mismatch. It remains useful as an interference control and cannot identify equal-compute semantic effects. Duplicate-label pairwise gaps are dependent and post-hoc. The task-level ranges are descriptive.

The update-proposal taxonomy used two language-model coders, one of which also generated the proposals it coded. Exact agreement on representational form is a cross-model consistency check and does not substitute for independent human coding. Interpretive labels showed lower exact agreement and remain exploratory. Cohen’s kappa is undefined for invariant fields because both coders have zero marginal variance; high exact agreement must not be read as a stable chance-corrected estimate. Configuration masking may be incomplete because proposal form can reveal substrate.

The retrieval analysis identifies lexical selection mechanics and does not establish causal harm to score. Offline re-ranking produced no new behavioral outcomes. No retrieval-harm experiment or consolidation intervention is part of this study.

Systematic resource summaries establish descriptive trajectory divergence, while the three complete-chain cases provide selected mechanism audits. The gated trace archive prevents independent inspection of those cases from the compact package. The cases are therefore illustrative, trace-dependent audits, and neither source identifies a general causal behavioral mediation effect.

The package records `gpt-5.6-sol` and `gpt-5.6-terra` as distinct coder identifiers and does not establish provider or pretraining independence. The outer endpoint’s sampling parameters and determinism guarantees are also not preserved in the compact package. The removed author-classified related-work matrix and all stronger comparative literature claims remain outside the paper’s evidence. The retained conceptual descriptions require independent primary-source verification before external circulation.

The local ActiveGraph benchmark and the Hybrid Pack share author-built technology. Sealed checks and an observed result unfavorable to Hybrid reduce a simple directional-bias concern but do not remove the structural conflict.

## 12 Conclusion

All three recorded configurations retained persistent changes. Their evaluation interface expressed those changes mainly as text, their controls exhibited substantial outcome spread and left interference uncontrolled, and none met the frozen uplift criterion. This is a descriptive result from one lineage per configuration against one labeled no-context draw, not a population null. Effects below the coarse family planning differences could not be reliably detected. The result separates five claims that are often compressed into one: retention, expression, behavioral mediation, task improvement, and recursive improvement.

The central lesson is operational. Treat reflection templates, acceptance gates, retrievers, adapters, execution paths, and controls as components of the self-improving system. Measure each link with receipts and matched interventions. In the studied system, the interface and acceptance mechanism constrained which retained modifications were admitted, expressed, and evaluated. This observed constraint does not quantify an effect magnitude or establish a universal interface law.

**Artifact availability.** The compact release can regenerate the manuscript, figures, tables, control sensitivities, audits, and manifest from included data. It supports inspection of the adjudication and frozen coding artifacts. Complete trajectories and trace-dependent case reproduc-

tion remain gated pending disclosure review and deposit; the DOI is pending. Unexercised control-ladder rungs and descendant productivity are proposed methods, not empirical outputs of this study.

## References

- [1] Eric Zelikman, Eliana Lorch, Lester Mackey, and Adam Tauman Kalai. Self-taught optimizer (STOP): Recursively self-improving code generation, 2023. <https://arxiv.org/abs/2310.02304>.
- [2] Xunjian Yin, Xinyi Wang, Liangming Pan, Xiaojun Wan, and William Yang Wang. Gödel agent: A self-referential agent framework for recursive self-improvement, 2024. <https://arxiv.org/abs/2410.04444>.
- [3] Maxime Robeyns, Martin Szummer, and Laurence Aitchison. A self-improving coding agent, 2025. <https://arxiv.org/abs/2504.15228>.
- [4] Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, and Jeff Clune. Darwin gödel machine: Open-ended evolution of self-improving agents, 2025. <https://arxiv.org/abs/2505.22954>.
- [5] Wenyi Wang, Piotr Piękos, Li Nanbo, Firas Laakom, Yimeng Chen, Mateusz Ostaszewski, Mingchen Zhuge, and Jürgen Schmidhuber. Huxley-gödel machine: Human-level coding agent development by an approximation of the optimal self-improving machine, 2025. <https://arxiv.org/abs/2510.21614>.
- [6] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning, 2023. <https://arxiv.org/abs/2303.11366>.

## A Adjudication sensitivity

The raw run contained 215 immediately valid rows and 13 rows requiring policy adjudication. The final dataset retained all 228 attempts. Categories were six official SWE task timeouts, six Terminal agent-budget timeouts with completed zero verifiers, and one ActiveGraph snapshot contradiction resolved by a sealed grader-only retry of the hash-identical submission.

Eight of nine frozen `evolved - cold ablation` contrasts are identical when raw invalid rows are scored zero and when the frozen adjudication decisions are applied. The exception is Hybrid ActiveGraph: the contrast moves from -26.0 points under raw-invalid-as-zero to -10.0 points after the hash-identical quota-scheduler regrade. It remains negative and the frozen conclusion is unchanged. A raw complete-case analysis gives +7.3 points for this cell because it drops the evolved invalid row. Complete-case contrasts are reported only as sensitivity analyses because dropping bounded failures changes the estimand and can favor arms with more invalid outcomes.

The study’s single adjudication judgment moved a score in the evolved arm’s favor, and the frozen conclusion remained null.

Machine-readable table: `data/generated/adjudication-sensitivity.csv`

## B Timeout taxonomy

**Table 5:** Timeout and adjudication taxonomy.

| Category                      | Count | Treatment                        | Interpretation                                                                               |
|-------------------------------|-------|----------------------------------|----------------------------------------------------------------------------------------------|
| Official SWE task timeout     | 6     | Scored zero                      | Official evaluation exceeded the frozen 3,600-second task limit.                             |
| Terminal agent-budget timeout | 6     | Scored zero                      | Agent exhausted 1,800 seconds and the completed verifier returned zero.                      |
| ActiveGraph snapshot re-grade | 1     | Hash-identical grader-only retry | Original grader contradicted a receipt-bound parsable submission; no model action was rerun. |

Timeouts remain outcomes under the bounded-agent estimand. The taxonomy separates agent budget exhaustion, official evaluator limits, and infrastructure contradiction.

## C Success-conditional efficiency

For the binary SWE and Terminal families, the companion table reports mean cost and wall time among successful attempts for every architecture and arm: `data/generated/success-conditional-efficiency.csv`.

These quantities describe the resource profile of observed successes. They do not estimate causal efficiency because conditioning on success selects different tasks and trajectories across arms. Unconditional costs remain the primary resource comparison.

## D Provenance

The study specification is `research/local_study.json`, SHA-256 `8313d21971452704893200d2022718fec985142ae32fe7d16932d34a0e0d581`, with frozen status at study commit `559843894578141dfbd39b7bae2246db76e57bde` dated `2026-07-17T18:30:38-07:00`. The compact release does not retain an independently witnessed timestamp for the first development call. The Round 1 manuscript package was commit `8bdff502cc08be968b21bbe5ebf177d49879a0fb`.

The provenance manifest records byte counts and SHA-256 hashes for the paper package’s frozen analytical datasets and reports, adjudication ledger, post-hoc metrics, masked coding packet, both original coder label files, agreement report, and manuscript: `provenance-manifest.json`.

The adjudication ledger separately hashes each changed row and its supporting evidence. Both original taxonomy-label files were frozen before adjudication.

## E Cross-model coding agreement

The table reports exact agreement, Cohen’s kappa, and Gwet’s AC1 for two separately executed, outcome-blind model-coding passes using the recorded identifiers `gpt-5.6-sol` and `gpt-5.6-terra`. Sol generated the 84 proposals it later coded; Terra supplied the other pass. The package does not establish provider or pretraining independence between these identifiers. The analysis is therefore a cross-model consistency check rather than independent human reliability. Kappa is undefined when both coders use one category for every item because the marginals have zero variance. AC1 is a supplemental prevalence-robust coefficient computed with the full pre-specified category vocabulary. Its value of 1.000 for invariant fields is mechanical and supplies no evidence beyond 84/84 exact agreement.

**Table 6:** Cross-model coding consistency. Sol generated the proposals it coded; undefined kappa indicates zero marginal variance.

| Field                           | Exact agreement | Cohen’s kappa | Gwet’s AC1 |
|---------------------------------|-----------------|---------------|------------|
| Primary update target           | 76/84 (90.5%)   | 0.000         | 0.904      |
| Secondary update target         | 47/84 (56.0%)   | -0.081        | 0.543      |
| Representational form           | 84/84 (100%)    | undefined     | 1.000      |
| Transfer scope                  | 63/84 (75.0%)   | 0.470         | 0.717      |
| Abstraction level               | 59/84 (70.2%)   | 0.529         | 0.647      |
| Evidence grounding              | 59/84 (70.2%)   | 0.428         | 0.669      |
| Failure specificity             | 83/84 (98.8%)   | 0.661         | 0.988      |
| Validation strategy             | 82/84 (97.6%)   | 0.592         | 0.976      |
| Consolidation operation         | 84/84 (100%)    | undefined     | 1.000      |
| Executable status               | 84/84 (100%)    | undefined     | 1.000      |
| Anticipated activation          | 84/84 (100%)    | undefined     | 1.000      |
| Counterfactual actionability    | 84/84 (100%)    | undefined     | 1.000      |
| Novelty relative to prior state | 84/84 (100%)    | undefined     | 1.000      |
| Confidence                      | 63/84 (75.0%)   | 0.393         | 0.687      |

For primary update target, Sol assigned 76 procedure and eight validation-asset labels, while Terra assigned all 84 to procedure. The resulting confusion is 76 procedure/procedure and eight validation-asset/procedure cases, which explains 90.5% exact agreement with kappa

0.000. Complete coder marginals and disagreements are retained in `data/taxonomy/coder-agreement.json`.

No coefficient interval is reported because these are two fixed model-coding passes used as a prompt and model-sensitivity check rather than sampled human raters. Interpretive fields remain exploratory.

## F Actor-visible control-collapse sensitivity

Source inspection confirms that cold and cold ablation use the same task, model, seed field, tools, workspace, system instructions, empty retained-context field, and later actor-visible tool protocol. Cold ablation validates the lineage state hash before suppressing its contents; that provenance difference never reaches the actor. The frozen `evolved - cold ablation` contrast is therefore retained as a decision record, not described as a separable lineage intervention.

The pooled-six and leave-the-matched-ablation-out calculations use only the committed compact score table. Across Workspace, Minimal, and Hybrid, the pooled-six deltas are +6.7, -3.3, and +6.7 points for SWE; +11.1, -22.2, and -5.6 for Terminal; and -9.9, +4.1, and -9.9 for ActiveGraph. The corresponding leave-one-out deltas are +8.0, -6.0, +6.0; +10.0, -20.0, -6.7; and -8.4, +3.2, -9.9 points.

The nominal 95% deterministic task-resampling intervals use 10,000 draws and the same percentile convention as the frozen comparison. For Workspace, Minimal, and Hybrid respectively, pooled-six intervals are SWE [-8.3, +26.7], [-30.0, +18.3], [-3.3, +21.7]; Terminal [0.0, +27.8], [-50.0, 0.0], [-25.0, +8.3]; and ActiveGraph [-33.3, +3.7], [-6.3, +18.7], [-33.3, +3.7]. Leave-one-out intervals are SWE [-10.0, +32.0], [-32.0, +16.0], [-4.0, +20.0]; Terminal [0.0, +30.0], [-46.7, 0.0], [-30.0, +10.0]; and ActiveGraph [-29.6, +4.4], [-5.6, +15.2], [-32.0, +2.4]. No lower bound is above zero, so the sensitivity analyses do not produce the required positive pattern in two families. The written no-uplift decision is unchanged.

A tie-aware conditional exchangeability calculation pools each evolved score with its six actor-visible-equivalent controls and uniformly relabels the focal score. Only a unique minimum or maximum is strictly outside the other six. Each case therefore contributes 0, 1/7, or 2/7 according to the number of unique extrema among its seven observed values. In these data, exactly 12 cases contribute 1/7 and the other 45 contribute zero. The contributors are: Workspace, Minimal, and Hybrid on SWE `pydata__xarray-3305`; Minimal and Hybrid on SWE `sympy__sympy-19040`; Minimal on SWE `scikit-learn__scikit-learn-13124`; Workspace and Hybrid on Terminal `bn-fit-modify`; Workspace, Minimal, and Hybrid on ActiveGraph `delegated_access_control`; and Minimal on ActiveGraph `quota_scheduler`.

Across 57 configuration-task cases, the expected

outside-range count is 12/7 (1.71) and the observed count is one. The expected counts are 1/7 among 36 cases on stable control tasks and 11/7 among 21 cases on variable control tasks. Under independent conditional relabelings across cases, the count variance is 72/49 (1.469). Cases share tasks and control observations, so that variance is a descriptive reference rather than a sampling-design claim. Machine-readable outputs are `data/generated/equivalent-control-sensitivity.csv` and `data/generated/outside-range-exchangeability.json`.